A-A-50966
30 March 1984

COMMERCIAL ITEM DESCRIPTION

CHUCKS, DRILL, AUTOMATIC, TAPER SHANK WITH TANG

The General Services Administration has authorized the use of this commercial item description in preference to the Type III and Type IV chucks covered by Federal Specification GGG-C-350.

This description covers automatic chucks which are designed to hold drills or other straight shank tools in drill presses, boring mills, jig borers, milling machines, motor shafts and other applications. The design and construction shall be such that tools can be safely and speedily changed while the spindle is rotating.

Salient characteristics.

Classification. The drill chucks covered by this description shall be of the following types and of the sizes specified in table I or table II for the type chuck required. The type and size chuck required for a specific requirement shall be as specified in the contract or order.

Type I - Three-jaw, mounted arbor with tapered shank
Type II - Four-jaw, integral arbor with tapered shank

Size. The size of the chucks shall be as listed in tables I or II for the type chuck required.

TABLE I. Sizes and capacities of type I chucks.

Size	Chuck capacity, inches	Shank taper, (ANSI)
1	15/64 to 1/2	1
2	15/64 to 1/2	2
3	15/64 to 1/2	3
4	3/8 to 3/4	2
5	3/8 to 3/4	3
6	3/8 to 3/4	4
7	17/32 to 1	3
8	17/32 to 1	4

FSC 3460

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TABLE II. Sizes and capacities of type II chucks.

Size	Chuck capacity, inches	Shank taper, (ANSI)
1	1/64 to 3/8	1
2	1/64 to 3/8	2
3	1/32 to 1/2	1
4	1/32 to 1/2	2
5	1/32 to 1/2	3

ANSI - American National Standards Institute

Design. The drill chucks covered by this CID shall be the latest design, easy to operate, and entirely adequate for drilling operations with drill presses and other type drilling and boring machines.

Material. The drill chucks shall be built of material entirely suitable for the purpose intended.

Chuck jaws. Type I chucks shall have three straight roller type jaws. Type II chucks shall have four tapered jaws. All chuck jaws shall have a Rockwell C scale hardness value of not less than 52.

Tapered shank. Type I and type II chucks shall have a tapered shank which shall conform to the requirements of table I and table II respectively, and shall meet the requirements specified in the latest issue of ANSI B5.10. The tapered shank shall be assembled and pinned to the type I chuck. The type II chuck shall have a built in solid taper shank. The shank tapers shall not be harden ed. The end of each tang shall have a Rockwell C scale hardness value of 40 to 55. The surface of the shank shall be free from wheel and chatter marks, nicks, cracks, or other imperfections which may affect accuracy, appearance or serviceability.

Concentricity. The tapered shank shall be ground concentric within 0.001 inch total indicator reading.

Shell and sleeve. The shell and sleeve of each drill chuck shall be knurled to provide a positive grip when changing drills.

Jaw shell. Jaw shells of the type II drill chucks shall be of the tapered bottom design for an unobstructed view of close tolerance work.

Operation. The chuck jaws shall be self-centering. The gripping power of the jaws shall automatically increase in direct ratio to strain on the enclosed tool without damaging the tool shank. The roller jaws of the type I drill chuck shall separate when the

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sleeve of the chuck is gripped. When the grip upon the sleeve is released, the jaws shall close. The jaws of the type II drill chuck shall open when the sleeve of the chuck is gripped and pulled down. Giving the tapered portion of the jaw shell an upward thrust shall close the jaws.

Markings. Each drill chuck shall be marked in a permanent and legible manner with the manufacturer's name, model number, capacity and taper.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for ther performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the CID where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples specified in table III that is applicable to the quantity of items being purchased. Each sample shall be visually inspected and tested by the supplier to determine conformance with all requirements specified herein. Any sample failing to meet these characteristics shall be considered a defective item. If the number of defective items equals, or is less than the quanity specified under "Acceptance, Number Defective" in table III, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, Number Defective" column on table III, the entire lot shall be considered defective.

TABLE III. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance number defective	Rejection number defective
8 and under	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	1	2
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
281 to 500	50	5	6

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Quality conformance inspection. Quality conformance inspection shall be applied to each sampled drill chuck prior to its being offered for acceptance under the contract or order. Quality conformance inspection shall consist of the following:

Operational test. Each of the sample drill chucks selected in accordance with table III shall be placed in a machine-tool spindle and tested for correct operational actions specified herein.

Concentricity test. Each sampled chuck shall be checked for concentricity specified herein.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes. The American National Standards Institute ANSI B5.10, "Machine Tapers", is available from the American National Standards Institute, Incorporated, 1430 Broadway, New York, NY 10018.

Purchasers should specify the quantity, type and size chuck required for a specific procurement.

Cross reference. The type III and type IV drill chucks in Federal Specification GGG-C-350 correspond to type II and type I respectively in this CID.

MILITARY INTERESTS:

Custodian

Air Force - 99

Review Activities

DLA - GS
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CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY

DLA - IP

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